

Skill-9

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In file included from /usr/include/x86_64-linux-gnu/bits/local_lim.h:38,
                 from /usr/include/x86_64-linux-gnu/bits/posix1_lim.h:161,
                 from /usr/include/limits.h:195,
                 from /usr/lib/gcc/x86_64-linux-gnu/13/include/limits.h:205,
                 from /usr/lib/gcc/x86_64-linux-gnu/13/include/syslimits.h:7,
                 from /usr/lib/gcc/x86_64-linux-gnu/13/include/limits.h:34,
                 from mini_shell1.c:5:
/usr/include/linux/limits.h:11: note: this is the location of the previous definition
11 | #define MAX_INPUT      255      /* size of the type-ahead buffer */
    |
=====
          MINI SHELL
=====
mysHELL$ cd /tmp
mysHELL$ pwd
/tmp
mysHELL$ cd /abc/xyz
cd: No such file or directory
mysHELL$ cd /home
mysHELL$ cd -
/tmp
mysHELL$ echo Hello
Hello
mysHELL$ hello
mysHELL: command not found: hello
mysHELL$ exit
Exiting shell...
himasree@LAPTOP-41COT5AA:~/OSLab$
```

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himasree@LAPTOP-41COT5AA:~/OSLab$ cat mini_shell1.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

#define MAX_INPUT 1024
#define MAX_ARGS 100

/* =====
   BUILT-IN COMMAND FUNCTIONS
   ===== */

int builtin_cd(char **args)
{
    char current_dir[PATH_MAX];
    char previous_dir[PATH_MAX];
    char *target;

    /* Save current directory */
    if (getcwd(current_dir, sizeof(current_dir)) == NULL) {
        perror("getcwd");
        return 1;
    }

    /* No argument -> go to HOME */
    if (args[1] == NULL) {
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    target = getenv("OLDPWD");

    if (target == NULL) {
        fprintf(stderr, "cd: OLDPWD not set\n");
        return 1;
    }

    printf("%s\n", target);
}

/* Normal path */
else {
    target = args[1];
}

/* Validate and change directory */
if (chdir(target) != 0) {
    perror("cd");
    return 1;
}

/* Save previous directory */
if (setenv("OLDPWD", current_dir, 1) != 0) {
    perror("setenv");
    return 1;
}

/* Update PWD */
if (getcwd(previous_dir, sizeof(previous_dir)) != NULL) {
    if (setenv("PWD", previous_dir, 1) != 0) {

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        perror("pwd");
        return 1;
    }

    printf("%s\n", cwd);

    return 1;
}

/* echo built-in */
int builtin_echo(char **args)
{
    for (int i = 1; args[i] != NULL; i++) {
        printf("%s", args[i]);

        if (args[i + 1] != NULL)
            printf(" ");
    }

    printf("\n");

    return 1;
}

/* exit built-in */
int builtin_exit(char **args)
{
    printf("Exiting shell...\n");

```

```

printf("=====\n");
printf("      MINI SHELL\n");
printf("=====\n");

while (1) {

    printf("myshell$ ");
    fflush(stdout);

    /* Read command */
    if (fgets(input, sizeof(input), stdin) == NULL)
        break;

    /* Remove newline */
    input[strcspn(input, "\n")] = '\0';

    /* Ignore empty input */
    if (strlen(input) == 0)
        continue;

    /* Convert input into tokens */
    tokenize(input, args);

    if (args[0] == NULL)
        continue;

    /* Check dispatch table */
    builtin_function command = find_builtin(args[0]);

```

